

Yongyi Jia 贾永奕

Ph.D. Student · Department of Automation · Tsinghua University

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Research Interests

Dexterous manipulation; arm–hand grasp generation; visual–tactile representation learning and VLA policies; robot learning, planning and control; microrobotics.

Education

Tsinghua University · Ph.D., Control Science and Engineering

2021.09–2027.06 (expected)

Department of Automation · Advisor: Prof. Xiang Li · GPA: 3.96/4.0

Tsinghua University · B.Eng., Automation

2017.09–2021.06

GPA: 3.6/4.0 · Top 20%

Research Experience

Huawei 2012 Laboratories · Research Collaboration

2025.01–2026.08

Human-prior grasp data generation (3.2M grasps, 157K scenes), arm-aware diffusion guidance, and tactile MPC for coordinated finger–wrist grasp execution.

Tsinghua University · Doctoral Research

2021–Present

Contact-aware object retrieval and in-hand manipulation; magnetic, optoelectronic and optical micromanipulation. Hierarchical RL, online dynamics learning, MPC, and physics-informed flow prediction.

Publications

1. Arm-Aware Guided Dexterous Grasp Generation with Arm-Agnostic Grasp Models

Yongyi Jia, Yongpeng Jiang, Kangchen Lv, Yi Ren, Xiang Li

IEEE RA-L, 2026. [Paper](#)

2. HUGS: Guiding Unified Dexterous Grasp Synthesis Across Modes and Scales via Learned Human Priors

Mingrui Yu*, Yongpeng Jiang*, Yongyi Jia, Kangchen Lv, Li Huang, Yi Ren, Xiang Li

CoRL, 2026. [Paper](#)

3. CoorGrasp: Coordinated Contact Control for Adaptive Dexterous Grasping Under Uncertainty

Mingrui Yu, Yongpeng Jiang, Yongyi Jia, Yi Ren, Xiang Li

ICRA · Oral, 2026. [Paper](#)

4. Kinematics-Aware Diffusion Policy with Consistent 3D Observation and Action Space for Whole-Arm Robotic Manipulation

Kangchen Lv, Mingrui Yu, Yongyi Jia, Chenyu Zhang, Xiang Li

IEEE RA-L, 2026. [Paper](#)

5. A Hybrid Modeling Method for Multi-Fingered Dexterous Robot Hands

Qikai Zou, Yongpeng Jiang, Yongyi Jia, Shu Miao, Zhixi Shen, Xiang Li

ICCA, 2026. [Paper](#)

6. **Robust In-Hand Reorientation With Hierarchical RL-Based Motion Primitives and Model-Based Regrasping**
Yongpeng Jiang, Mingrui Yu, Chen Chen, Yongyi Jia, Xiang Li
IEEE RA-P, 2025. [Paper](#)
7. **Robotic In-Hand Manipulation for Large-Range Precise Object Movement: The RGMC Champion Solution**
Mingrui Yu, Yongpeng Jiang, Chen Chen, Yongyi Jia, Xiang Li
IEEE RA-L, 2025. [Paper](#)
8. **Contact-Aware Non-Prehensile Manipulation for Object Retrieval in Cluttered Environments**
Yongpeng Jiang*, Yongyi Jia*, Xiang Li
IROS, 2023. [Paper](#)
9. **Non-contact Dexterous Micromanipulation with Multiple Optoelectronic Robots**
Yongyi Jia, Shu Miao, Ao Wang, Caiding Ni, Lin Feng, Xiaowo Wang, Xiang Li
IEEE RA-L, 2025. [Paper](#)
10. **Flow-Aware Navigation of Magnetic Micro-Robots in Complex Fluids via PINN-Based Prediction**
Yongyi Jia, Shu Miao, Jiayu Wu, Ming Yang, Chengzhi Hu, Xiang Li
IROS, 2025. [Paper](#)
11. **Particle-Assisted Optoelectronic Tweezers for Manipulating Single Cells and Microparticles**
Ao Wang, Shuzhang Liang, Caiding Ni, Yongyi Jia, Kangjie Wu, Wenyan Niu, Shunxiao Huang, Kaiyi Peng, Chutian Wang, Yingjian Guo, Zhijun Zhao, Lingze Zhang, Mingjie Liu, Lin Feng
Advanced Science, 2025. [Paper](#)
12. **Efficient Model Learning and Adaptive Tracking Control of Magnetic Micro-Robots for Non-Contact Manipulation**
Yongyi Jia, Shu Miao, Junjian Zhou, Niandong Jiao, Lianqing Liu, Xiang Li
ICRA, 2024. [Paper](#)
13. **Cell Cryopreservation in a Microfluidic Chip with Vision-Based Fluid Control and Region Reaching**
Shu Miao, Yongyi Jia, Ze Jiang, Jiehuan Xu, Xiang Li
IEEE T-ASE, 2024. [Paper](#)
14. **Hierarchical Learning and Control for In-Hand Micromanipulation Using Multiple Laser-Driven Micro-Tools**
Yongyi Jia, Yu Chen, Hao Liu, Xiu Li, Xiang Li
IROS, 2022. [Paper](#)
15. **Transblast-Net: Semantic Segmentation of Human Blastocyst Components with Composed CNN and Transformer Architecture**
Shu Miao, Yongyi Jia, Xiang Li
RCAR, 2025. [Paper](#)
16. **高危化工环境机器人应用研究进展**
李翔, 于铭瑞, 贾永奕, 吕康晨, 宋士吉
新兴科学技术趋势, 2022. [Paper](#)
17. **高危化工机器人研究与应用综述**
于铭瑞, 贾永奕, 李翔
中国自动化大会 · CAC, 2021. [Paper](#)

Awards & Honors

- Best Poster Award · ICRA Workshop on Dexterity with Multifingered Hands (2026)
- China National Scholarship for Doctoral Students (2025)
- Best Medical Robotics Paper Award Finalist · ICRA (first author) (2024)
- Champion, In-Hand Manipulation & Most Elegant Solution · ICRA RGMC (2024)
- Second Place · ICRA Robotic Simulation Manipulation Challenge (2023)
- Comprehensive Excellence Scholarship · Tsinghua University (2023–24)

Technical Experience

PyTorch; VLA training and fine-tuning; multimodal representation learning; diffusion policies; reinforcement learning; motion planning; model predictive control; teleoperation data collection; simulation and real-robot deployment.

Academic Service

Reviewer: IEEE T-RO, RA-L, T-ASE, ICRA, and IROS.

Vice President, Student Science & Technology Association; Head of Academic Affairs, Department of Automation, Tsinghua University.

Updated October 2026. Selected publication list; linked project and paper versions may differ.